

5M
Q. Discuss the Role of Growing-Population, Industrialization, Urbanization & Changing lifestyles in contributing to Environmental degradation. What measures can be taken to mitigate their impact?

Q. What is ^{by GUIAC} Environmental Degradation? Causes and Mitigation

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Environmental degradation refers to the **deterioration of the natural environment** due to human activities, leading to issues like pollution, climate change, deforestation, and loss of biodiversity. Several key factors contribute significantly to this problem:

It Causes: ~~are~~ due to following factors:

1. Growing Population

- **Impact:**
 - Increases demand for land, water, food, and energy.
 - Leads to deforestation, overuse of natural resources, and waste generation.
 - More people = more pressure on limited ecosystems.
- **Example:** Rapid population growth in cities like Delhi or Lagos has led to severe air and water pollution, and unmanageable waste.

2. Industrialization

- **Impact:**
 - Industries release toxic chemicals, greenhouse gases, and effluents.
 - Promotes unsustainable resource extraction (e.g., mining, fossil fuels).
 - Often disregards ecological balance for economic growth.
- **Example:** Industrial zones like Bhiwandi (India) or Wuhan (China) have faced repeated air and water pollution issues.

3. Urbanization

- **Impact:**
 - Expansion of cities leads to loss of green cover and wetlands.
 - Increases traffic congestion, waste, and air pollution.
 - Creates heat islands and puts pressure on sewage, water, and power systems.
- **Example:** In cities like Mumbai, unregulated urban growth has led to frequent flooding and habitat loss.

4. Changing Lifestyle

- **Impact:**
 - Shift towards consumerism and high energy consumption.
 - Increase in single-use plastics, e-waste, fast fashion, and processed food.
 - Overreliance on vehicles, gadgets, and packaged goods worsens pollution.
- **Example:** Fast fashion industries contribute to massive textile waste and water pollution, especially in developing countries.

✓ Measures to Mitigate the Impact

Policy and Planning:

- Implement strict environmental laws and pollution control norms. (policies)
- Promote sustainable urban planning (green buildings, public transport).

Public Awareness and Education:

- Promote **eco-friendly lifestyles** (reduce, reuse, recycle).
- Incorporate **environmental education** in school curricula.

Technology and Innovation:

- Use **clean energy sources** (solar, wind, hydro).
- Adopt **waste management technologies** like composting and recycling.

Population Control:

- Promote **family planning** and **women's education**.
- Strengthen health and reproductive services.

Corporate Responsibility:

- Enforce **green auditing** and sustainability reporting in industries.
- Encourage **CSR initiatives** focusing on the environment.

Conclusion:

To ensure a healthy planet for future generations, it is crucial to **balance development with environmental conservation**. Collective efforts from governments, industries, and individuals are essential to combat the growing threats of environmental degradation.

Combat = ^{try to} Stop, ^(stop) to fight-against, try to defeat

Perceived → Noticed / viewed / understand or think in particular -
 ↳ expected / Noted - understood.
 ↳ viewed = Looked at - / observed - or specified way.
 ↳ distinguished / inspected

5M

Q. Explain in detail about "paradigm shift in disaster management"

[10 marks]

The "paradigm shift in disaster management" refers to a fundamental change in the way disasters are perceived, planned for, and managed. Traditionally, disaster management focused mainly on responding after a disaster occurred. But over the past few decades, there has been a major shift toward a more proactive, comprehensive, and integrated approach, emphasizing preparedness, mitigation, risk reduction, and resilience. ^{Return} [ability to recover quickly to a good condition] after facing the problem.

Traditional Approach vs Modern Approach (Paradigm Shift)

Various Aspects	Traditional Disaster Management	* Modern Disaster Management (Paradigm Shift) [in Advance]
1. Focus	Response and Relief	Risk Reduction, Preparedness, and Resilience
2. Timing	Post-Disaster	Pre-, During, and Post-Disaster
3. Approach	Reactive	Proactive and Preventive
4. Responsibility	Primarily Government	Multi-Stakeholder (Govt., NGOs, Community, Private Sector)
5. Planning	Crisis-driven	Risk-informed and Continuous approach.
6. Community Involvement	Passive victims	Active participants in planning and response
7. Integration with Development	Separate from development planning	Integrated with sustainable development goals (SDGs)
8. Objective	Save lives during disaster	Build a disaster-resilient community/society

Key Elements of the Paradigm Shift

- From Relief to Risk Reduction**
 - Earlier, focus was on providing emergency relief like food, shelter, and medical aid after a disaster.
 - Now, the focus is on identifying risks and reducing vulnerability through mitigation and preparedness.
- Community Participation**
 - Communities are no longer seen as helpless victims.
 - They are now seen as the **first responders** and **key players** in disaster preparedness and recovery.
- Integration with Development**
 - Disaster risk reduction is now embedded into **urban planning, infrastructure design, agriculture, education**, etc.
 - Example: Building earthquake-resistant houses or flood-proof roads.
- Use of Technology**
 - * Modern disaster management relies heavily on **GIS, satellite imagery, early warning systems**, and **AI** for better preparedness and timely response.

Crisis: At a time of great danger or difficulty (संकट काल)

Ex: Financial Crisis → [Condition of instability or danger]

Risk informed approach = ^{Way of} Making decisions by considering potential risks & their possible (in advance) consequences

5. Institutional Changes

- Countries have created **dedicated disaster management authorities** (e.g., NDMA in India) and legal frameworks.
- Policies emphasize **disaster risk governance** at all levels.

6. Building Resilience

- Instead of just bouncing back after a disaster, the aim is to **bounce forward** — becoming stronger and more adaptable.

■ Global Frameworks Supporting This Shift

- **Hyogo Framework for Action (2005–2015)** – Emphasized building the resilience of nations and communities.
- **Sendai Framework for Disaster Risk Reduction (2015–2030)** – Prioritizes understanding risk, strengthening governance, investing in resilience, and enhancing preparedness.

IN Example: Paradigm Shift in India

- India moved from a **relief-centric approach** (pre-2000) to a **proactive disaster risk management approach** post-2005 with the formation of the **National Disaster Management Authority (NDMA)**.
- Programs like the **National Cyclone Risk Mitigation Project**, **School Safety Program**, and **Smart Cities Mission** integrate disaster risk reduction into planning.

★ Conclusion

The **paradigm shift in disaster management** is about **anticipating disasters, reducing risks, and building resilient communities** rather than just reacting after a disaster strikes. It reflects a **holistic, inclusive, and development-oriented approach**, aligning closely with global goals like the **SDGs and climate action frameworks**.